

NETWORKS AND PROTOCOLS

Exercises Sheet 6

Exercise 1

Part 1

Answer the following questions with **yes**, **no**, or **maybe**.

1. The website `google.dz` is hosted in Algeria
2. The website `google.dz` can be hosted outside Algeria
3. The website `google.dz` can be hosted inside and outside Algeria

Part 2

1. We run the following command `dig example.edu` from a terminal. Fill in the gaps (represented by ...) in the following output and answer subsequent questions:

```
;; QUESTION SECTION:
```

```
; ...                IN          A
```

```
;; ANSWER SECTION:
```

```
...                86400      IN          A          93.184.216.119
```

```
;; AUTHORITY SECTION:
```

```
...                172800    IN          NS          b.iana-servers.net.
```

```
...                172800    IN          NS          ...
```

```
;; ADDITIONAL SECTION:
```

```
a.iana-servers.net. 22728    IN          A          199.43.132.53  
...                22728    IN          AAAA         2001:500:8c::53  
...                22728    IN          A          199.43.133.53  
...                22728    IN          AAAA         2001:500:8d::53
```

2. Does the host `b.iana-servers.net` have an IPv6 address? If so give it, if not justify.
3. What does `93.184.216.119` represent?
4. What are the name(s) of the domain name servers that have authority over the domain `example.edu`?

Exercise 2

1. Recall the role of a DNS server.
2. A web browser is opened at the URL below:
`http://www.iut.edu/rep/file.html`
 - (a) What service protocols will the browser use?
 - (b) What messages will be exchanged between our machine and the other servers involved?
3. Assume that the DNS server used by the machine is down. How can the user access the file `/rep/file.html`?
4. Consider the following excerpt from the DNS zone file for the domain `iut.edu`, which is hosted on the server `dns.iut.edu`. The file contains records in the following format:

`<name> <type> <value>`

<code>iut.edu.</code>	<code>NS</code>	<code>dns</code>
<code>dns</code>	<code>A</code>	<code>10.1.0.1</code>
<code>www</code>	<code>A</code>	<code>10.1.0.2</code>
<code>mail</code>	<code>A</code>	<code>10.1.0.3</code>
<code>max.student</code>	<code>A</code>	<code>10.2.0.100</code>
<code>anna.student</code>	<code>A</code>	<code>10.2.0.200</code>

Here, `<name>` identifies a machine or domain, which can be absolute (ending with a dot) or relative (relative to the zone name). Multiple records can be associated with a name, differentiated by their `<type>`. The `<value>` of the record depends on its type. In this example, there are two types: `A` for defining an IP address and `NS` (Name Server) for defining a DNS server for a zone.

- (a) What are the full names of the machines in the zone `iut.edu` for which the server knows the IP addresses?
- (b) We want the domain `student.iut.edu` to be delegated to another DNS server, called `dns.student.iut.edu` with IP address `10.2.0.1`. To clarify, our goal is to partition the zone `iut.edu` and establish a dedicated subzone for `student.iut.edu`. What steps are required to achieve this?
- (c) What will be the response of the server `dns.iut.edu` if it is queried about the machine `anna.student.iut.edu`?

Exercise 3

1. Describe the parameters that a DHCP server can provide to a client.
2. To ensure seamless recovery after an unexpected shutdown, such as power outage, what critical data must a DHCP server persistently store on disk?
3. Given that a DHCP client attempts to reclaim its previous IP address upon restarting, what are the potential outcomes and implications if the client is shutdown and then restarted?
4. In a network with multiple DHCP servers, how does a DHCP server determine that its offer for a client was not accepted?
5. Consider the following messages:

No.	Time	Source	Destination	Protocol	Info
1	0.000000	0.0.0.0	255.255.255.255	DHCP	DHCP Discover - Transaction ID 0X7e2c562a
2	0.017351	192.168.1.1	192.168.1.2	DHCP	DHCP Offer - Transaction ID 0X7e2c562a
3	0.017722	0.0.0.0	255.255.255.255	DHCP	DHCP Request - Transaction ID 0X7e2c562a
4	0.052182	192.168.1.1	192.168.1.2	DHCP	DHCP ACK - Transaction ID 0X7e2c562a
5	0.115033	00:00:61:C9:28:31	Broadcast	ARP	who has 192.168.1.27 Tell 192.168.1.2
6	0.610324	00:00:61:C9:28:31	Broadcast	ARP	who has 192.168.1.27 Tell 192.168.1.2
7	1.610364	00:00:61:C9:28:31	Broadcast	ARP	who has 192.168.1.27 Tell 192.168.1.2

- a. Identify the IP address of the DHCP server.
- b. Identify the IP address that the server proposes for the client.
- c. From message 1, does the client know the IP address of the DHCP server? Explain.
- d. From message 3, does the client know the IP address of the DHCP server? Explain.
- e. What is the purpose of the 3 ARP packets?

Exercise 4

Consider the configuration depicted in the figure above and answer the following questions.

1. Provide an IP configuration for the following hosts: (1) home machine A, (2) home router X, (3) Algeria Telecom DNS server, (4) university machine B, (5) university DHCP server, (6) university Web server, (7) university Email server.
2. How many machines can we connect to University Network?
3. Answer the following questions with Yes or No and justify your answer.
 - a) Machine A can ping Machine B.
 - b) Machine B can ping Machine A
 - c) Machine A can ping google.com.
 - d) google.com can ping Machine A.
 - e) Machine B can ping DHCP server.
 - f) Machine A can ping Email server.

